

Aden Walsey

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Master of Engineering in Mechanical Engineering, Early Admit

Bachelor of Science in Mechanical Engineering, Dean's List, Varsity Rowing, **GPA: 3.4**

Expected December 2026

Expected May 2026

SKILLS

Manufacturing: DFM/DFA, GD&T, CNC Machining, Manual Machining, Rapid Prototyping, QFD, BOMs, Fixture Design, Production Processes, Lean Manufacturing, Six Sigma.

CAD & Simulation: SolidWorks, Autodesk Fusion 360, ANSYS (CFD/FEA).

Analysis & Programming: MATLAB/Simulink, Python, R, C++.

Leadership: Cornell Varsity Lightweight Rowing: Second Team All-Ivy (2025), Head of the Charles 1st place in 1V4+ (2023); member of Cornell's Big Red Leadership Institute.

PROFESSIONAL EXPERIENCE

Manufacturing Engineer Intern, *Mainspring Energy*, Menlo Park, CA

Summer 2025

- Designed, assembled, and validated a pneumatic tooling jig to strip copper wire insulation and activate bond coat using electrical current, enabling stator coil formation and reducing production process time by 60%.
- Devised and executed production qualification test procedures for injection-molded shipping crates under simulated shipping conditions, verifying structural robustness and safety factors for secure delivery of stator assemblies.
- Conducted a Gage R&R analysis for eddy current sensors to quantify repeatability and reproducibility of precise alignment prior to stator shipment while ensuring efficiency of sensor installation.

R&D Engineer Intern, *Neodyne Biosciences, Inc.*, Fremont, CA

Summer 2024

- Developed and executed test methods to resolve adhesion failures, reducing production defects by over 70%.
- Performed tolerance analysis on device packaging to improve manufacturability and streamline assembly, creating detailed technical drawings in SolidWorks to document and communicate design improvements to manufacturing.

Robotics Teacher, *Pinewood School*, Los Altos, CA

Summer 2023

- Taught students how to design, build, and program robotics systems to develop their problem-solving skills.

PROJECTS

Simulations-Based Bicycle Frame Design and Analysis, *Advisor Rajesh Bhaskaran*, Cornell University

Spring 2026

- Design a bicycle frame for a community bike-share program in Fusion 360, focusing on manufacturable aluminum tube geometries, structurally sound frame architecture, and practical fabrication constraints for production frame assemblies.
- Construct a Finite Element Analysis (FEA) workflow in ANSYS to evaluate frame stiffness under pedaling loads, defining bottom bracket stiffness as the primary performance metric used in frame evaluation.
- Simulate and iterate tube geometries and wall thicknesses to quantify frame deformation, analyze structural load paths through the frame, and improve stiffness-to-weight performance.

Child Restraint System Capstone Project, *Advisor Marty Murtagh*, Cornell University

Fall 2025

- Led mechanical design of a modular, multi-stage child car seat platform in Autodesk Fusion, reducing part count and consolidating 3-4 traditional products into a single, modular system that spans from ages 6 months to 12 years old.
- Translated user and safety requirements into engineering targets using QFD and DFMEA, guiding material selection, structural layout, and failure-mode mitigation early in development.
- Validated structural performance with ANSYS under FMVSS 213 frontal crash load cases, achieving acceptable factors of safety with a final system weight under 20 pounds and 48% improved lifecycle sustainability.

Load Cell Finite Element Analysis, *Advisor Alan Zehnder*, Cornell University

Fall 2024

- Modeled a ring-shaped load cell with strain gauges in Fusion 360 to evaluate stress, strain, and deflection in ANSYS.
- Refined design parameters to maximize load cell's voltage output while confirming compliance with safety factors.
- Validated simulation results with hand calculations to ensure accuracy in measured stress distributions and properties.

INTERESTS

Cycling, Running, Backpacking, Cooking, Motorsport, Drawing, Hi-Fi Audio Systems, Furniture Design, Film Photography.